AI in Soil Moisture Prediction

When Soil Starts Predicting Its Own Thirst! 🌱🧠

Presented by: Yashwanth GP | **Student ID:** AMB2310

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The Thirsty Farmer 🚨

Traditional Guesswork

Manual checks unreliable, weather unpredictable

Consequences

Overwatering wastes resources, underwatering kills crops

Imagine if soil could text the farmer: "Hey, I'll be thirsty tomorrow at 2 PM!" 📱

Research Question: Can AI predict future soil moisture levels to optimise irrigation?

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Our Methodology 🔬

The farmer hires an AI Assistant that studies past soil behaviour like a topper preparing for exams 📚

01

Data Collection

IoT sensors continuously monitor soil and weather conditions

02

Data Preprocessing

Clean, validate, and prepare time-series sensor readings

03

Model Training

Apply ML algorithms to learn moisture patterns

04

Validation

Test predictions against historical records

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Data: The Foundation 📊

Soil Moisture

Current and historical readings from sensor networks

Temperature

Air and soil temperature variations throughout day

Humidity

Atmospheric moisture levels affecting evaporation

Precipitation

Rainfall patterns and intensity measurements

Irrigation

Watering schedules and application volumes

This is like tracking when the soil drinks water, sweats it out, or gets surprise rain showers! 🌧️

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The AI Soil Fortune Teller 🔮

Regression Models

Linear and polynomial relationships between variables

Time-Series Forecasting

ARIMA, LSTM networks for sequential patterns

Machine Learning

Random Forest, Gradient Boosting for accuracy

Goal: Predict soil moisture in advance, automate irrigation, reduce human intervention

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Results That Make Farmers Smile 😊

30%

Water Savings

Reduced irrigation without crop compromise 💧

↑

Accuracy

Significantly improved prediction reliability

↑

Crop Yield

Better health through precise watering 🌾

The farmer finally relaxes—no more overwatering disasters or dry crops!

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Transforming Farming Forever 🌱

Smart Farming

AI transforms traditional practices into data-driven precision agriculture

Reduced Uncertainty

Predictive models replace guesswork with confidence

Sustainability

Efficient water management protects resources for future generations

The lazy farmer becomes a smart farmer—thanks to AI doing the thinking! 🤖

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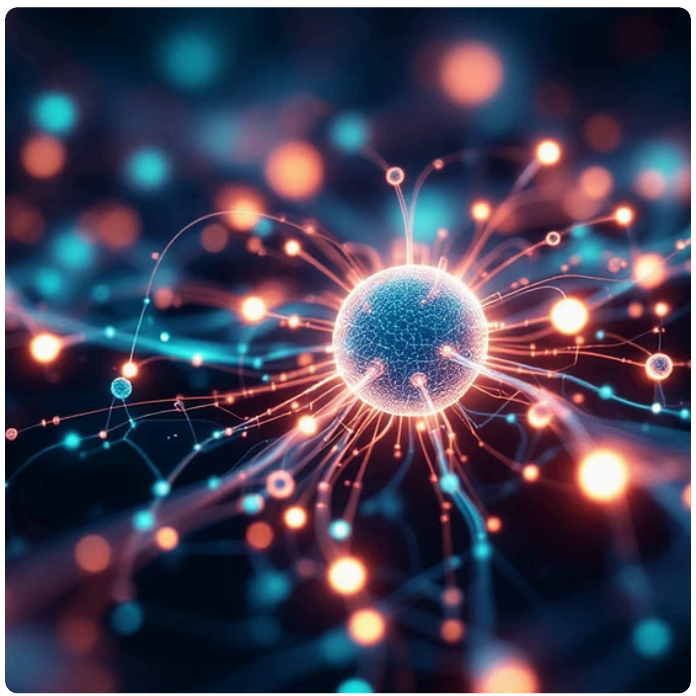
Future Possibilities 🚀

Aerial Monitoring

Drones supplement ground sensors with overhead data

Mobile Alerts

Real-time notifications when action needed



Deep Learning

More sophisticated models for higher accuracy

Discussion: If your soil could predict the future, what else would you want it to tell you?

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Thank You! 🙌

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> Let's make farming intelligent—because even soil deserves a smart future 🌱🤖

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